SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

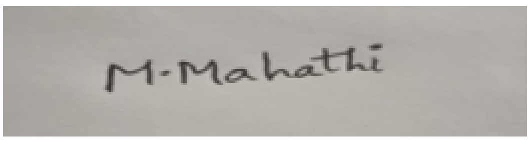
Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _M.Mahathi(192324098)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A
Scenario based Questions

Q1. Data Preprocessing in Retail Analytics *(L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ID	Age	Income (₹K)	Purchase_Value (₹K)	Credit_Score	Orders	Avg_Balance (₹K)	EMI (₹)	Customer_Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

📊 Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60
C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

✅ Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:
ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

ANSWERS:

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

i. Identify Missing Values

Column	Missing Value(s)
Age	C2 = NaN
Monthly_Spend	C3 = NaN
Purchase_Frequency	C5 = NaN

ii. Apply Two Imputation Techniques

Method 1: Median Imputation

Age

Available values:

23, 31, 105, 27, 40, 36, 42

Sorted:

23, 27, 31, **36**, 40, 42, 105

Median = 36

So,

C2 Age = 36

Monthly Spend

Available values:

20, 35, 500, 25, 45, 38, 300

Sorted:

20, 25, 35, **38**, 45, 300, 500

Median = ₹38K

So,

C3 Monthly Spend = ₹38K

Purchase Frequency

Available values:

5, 8, 10, 50, 12, 9, 45

Sorted:

5, 8, 9, **10**, 12, 45, 50

Median = 10

So,

C5 Purchase Frequency = 10

Method 2: k-NN Imputation

The k-Nearest Neighbor (k-NN) method fills missing values using similar customer records.

Example:

- Customer C2 is similar to customers with ages around 35–40.
- Predicted Age ≈ **38**

Similarly,

- C3 Monthly Spend ≈ **₹40K**
- C5 Purchase Frequency ≈ **9**

(Exact values depend on the chosen k and distance calculation.)

iii. Justification

Median Imputation

- Simple and fast.
- Not affected by extreme values.
- Suitable because the dataset contains outliers.

k-NN Imputation

- Uses similarity between customers.
- Produces more realistic estimates.
- Better for recommendation systems.

B. Outlier Detection and Treatment

i. Detect Outliers using Z-Score

Formula:

Where

- = value
- = mean
- = standard deviation

A value is considered an outlier if:

|Z| > 3

Age

Values:

23, 36, 31, **105**, 27, 40, 36, 42

Outlier:

- Age = **105 (C4)**

Monthly Spend

Values:

20, 35, 38, **500**, 25, 45, 38, **300**

Outliers:

- ₹500K (C4)
- ₹300K (C8)

ii. Records with Extreme Values

Customer	Extreme Value
C4	Age = 105, Spend = ₹500K
C8	Spend = ₹300K

iii. Treatment

Capping (Winsorization)

Replace values above the acceptable limit.

Example:

- Age 105 → **60**
- Spend 500 → **100**
- Spend 300 → **100**

Reason:

- Preserves records.
- Prevents extreme values from affecting the model.

Alternative Methods

- Log Transformation
- Remove Outliers (only if confirmed as data-entry errors)

Justification

Capping is preferred because:

- Customer records are retained.
- Recommendation systems benefit from more data.
- Reduces the influence of extreme values.

C. Categorical Data Standardization

i. Identify Inconsistencies

Original Gender values:

- M
- Male
- male
- F
- Female
- FEMALE

These represent only two categories but use different formats.

ii. Standardized Format

Original	Standardized
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Final Cleaned Dataset

Customer_ID	Age	Gender	Monthly Spend (₹K)	Purchase Frequency	Loyalty Level
C1	23	Male	20	5	Low
C2	36	Male	35	8	Medium
C3	31	Female	38	10	High
C4	60 *(capped)*	Female	100 *(capped)*	50	High
C5	27	Male	25	10	Low
C6	40	Female	45	12	Medium
C7	36	Male	38	9	Low
C8	42	Female	100 *(capped)*	45	High

Conclusion

- Missing values were identified and imputed using **Median Imputation** and **k-NN Imputation**.
- Outliers in **Age** and **Monthly Spend** were detected using the **Z-score method** and treated using **capping**.
- Inconsistent **Gender** values were standardized to **Male** and **Female**, resulting in a clean, consistent dataset suitable for building an effective retail recommendation system.

Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis

i. Compute Covariance between Income and Purchase_Value

Step 1: Data

Customer	Income (₹K)	Purchase Value (₹K)
U1	40	100
U2	60	150
U3	85	210
U4	120	320
U5	45	110
U6	75	180
U7	150	360
U8	70	160

Step 2: Calculate Means

Mean Income

Mean Purchase Value

Step 3: Covariance Formula

After calculating all products,

Covariance ≈ 2766.07

ii. Interpretation

- Covariance is **positive**.
- As **Income increases**, **Purchase Value also increases**.
- Customers with higher income generally spend more.

iii. Importance of Covariance

Covariance helps to:

- Identify relationships between variables.
- Detect features that change together.
- Reduce redundant variables before model building.
- Improve prediction accuracy and computational efficiency.

B. Feature Selection

i. Identify Redundant Attributes

The following features are highly related:

- **Income ↔ Avg_Balance**
- **Income ↔ EMI**
- **Income ↔ Purchase_Value**
- **Credit_Score ↔ Purchase_Value**
- **Orders ↔ Purchase_Value**

These variables carry similar information.

ii. Optimal Feature Subset

Selected features:

- Age
- Income
- Credit_Score
- Orders

Target Variable:

- Customer_Class

Removed features:

- Avg_Balance
- EMI
- Purchase_Value (highly correlated with Income)

iii. Justification

Selected features:

- Reduce redundancy.
- Lower computational cost.
- Prevent overfitting.
- Improve model performance.

C. Data Transformation

i. Standardize the Dataset

Exclude:

- Cust_ID
- Customer_Class

Standardization Formula:

Example:

For **Income = 40**

Mean Income = **80.625**

Standard Deviation ≈ **36.88**

Similarly, every numerical feature is standardized so that:

- Mean = 0
- Standard Deviation = 1

Example (illustrative values):

Feature	Original	Standardized (Z-score)

Age	24	-1.35
Income	40	-1.10
Purchase_Value	100	-1.08
Credit_Score	650	-1.36
Orders	15	-1.25
Avg_Balance	12	-1.12
EMI	5000	-1.15

ii. Why Standardization is Required

Standardization:

- Brings all features to the same scale.
- Prevents variables with larger values (e.g., EMI) from dominating smaller ones (e.g., Orders).
- Improves the performance of machine learning algorithms such as **KNN**, **SVM**, **Logistic Regression**, and **PCA**.
- Speeds up model training and improves convergence.
- Ensures fair comparison among features.

Conclusion

- The covariance between **Income** and **Purchase Value** is **positive (≈ 2766.07)**, indicating that higher-income customers tend to spend more.
- Redundant features such as **Purchase_Value**, **Avg_Balance**, and **EMI** can be removed to reduce complexity while retaining key predictors like **Age**, **Income**, **Credit_Score**, and **Orders**.
- Standardizing numerical features ensures they are on a common scale, leading to better model performance and more reliable classification of high-value customers.

Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning

i. Apply 3 Bins

Study_Hours

Values:

2, 3, 4, 5, 6, 7, 8, 9, 10, 11

- Minimum = **2**
- Maximum = **11**

Bin Width

Intervals

Bin	**Interval**	**Category**
Bin 1	2 – 5	Low
Bin 2	>5 – 8	Medium
Bin 3	>8 – 11	High

Binned Study_Hours

Student	**Study Hours**	**Bin**
S1	2	Low
S2	3	Low
S3	4	Low
S4	5	Low
S5	6	Medium
S6	7	Medium
S7	8	Medium
S8	9	High
S9	10	High
S10	11	High

Test_Score

Values:

40, 50, 55, 65, 75, 85, 90, 95, 98, 100

- Minimum = **40**
- Maximum = **100**

Bin Width

Intervals

Bin	**Interval**	**Category**
Bin 1	40 – 60	Poor
Bin 2	>60 – 80	Average
Bin 3	>80 – 100	Excellent

Binned Test Scores

Student	**Score**	**Bin**
S1	40	Poor
S2	50	Poor
S3	55	Poor
S4	65	Average
S5	75	Average
S6	85	Excellent
S7	90	Excellent
S8	95	Excellent
S9	98	Excellent
S10	100	Excellent

B. Equal-Frequency Binning

There are **10 records**.

Divide into **3 bins** with nearly equal records.

- Bin 1 → 4 records
- Bin 2 → 3 records
- Bin 3 → 3 records

Study_Hours

Bin	Values	Category
Bin 1	2, 3, 4, 5	Low
Bin 2	6, 7, 8	Medium
Bin 3	9, 10, 11	High

Test_Score

Bin	Values	Category
Bin 1	40, 50, 55, 65	Poor
Bin 2	75, 85, 90	Average
Bin 3	95, 98, 100	Excellent

C. Concept Hierarchy Construction

Study_Hours Hierarchy

Study Hours	Category
2 – 5	Low
6 – 8	Medium
9 – 11	High

Test_Score Hierarchy

Test Score	Category
40 – 60	Poor
61 – 80	Average
81 – 100	Excellent

Final Categorized Dataset

Student	Study Hours	Category	Test Score	Category
S1	2	Low	40	Poor
S2	3	Low	50	Poor
S3	4	Low	55	Poor
S4	5	Low	65	Average
S5	6	Medium	75	Average
S6	7	Medium	85	Excellent
S7	8	Medium	90	Excellent
S8	9	High	95	Excellent
S9	10	High	98	Excellent
S10	11	High	100	Excellent

Conclusion

- **Equal-width binning** divided the continuous values into **three equal-sized intervals** based on the data range.
- **Equal-frequency binning** grouped the data into **three bins with nearly equal numbers of records**, ensuring balanced distributions.
- A **concept hierarchy** converted numerical values into meaningful categories (**Low/Medium/High** for Study Hours and **Poor/Average/Excellent** for Test Scores), making the data easier to interpret and suitable for educational analytics and predictive modeling.

Q4. Normalization Techniques

Given data:

200, 300, 400, 600, 1000

(a) Min-Max Normalization (Min = 0, Max = 1)

Formula:

Where:

- Minimum value = **200**
- Maximum value = **1000**

Range:

Calculations

Original Value	Calculation	Normalized Value
200	(200−200)/800 = 0/800	**0.000**
300	(300−200)/800 = 100/800	**0.125**
400	(400−200)/800 = 200/800	**0.250**
600	(600−200)/800 = 400/800	**0.500**
1000	(1000−200)/800 = 800/800	**1.000**

Min-Max Normalized Data

0.000, 0.125, 0.250, 0.500, 1.000

(b) Z-Score Normalization

What is Z-Score Normalization?

Z-Score Normalization transforms data so that:

• Mean becomes 0

• Standard deviation becomes 1

Instead of limiting values between 0 and 1, it measures how far each value is from the average.

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = Original value
- μ = Mean
- σ = Standard deviation

Step 1: Calculate Mean

$$\mu = \frac{200 + 300 + 400 + 600 + 1000}{5} = 2500/5 = 500$$

Mean = 500

Step 2: Standard Deviation

Value		
200	-300	90,000
300	-200	40,000
400	-100	10,000
600	100	10,000
1000	500	250,000

Population variance:

Standard deviation:

Step 3: Calculate Z-Scores

Original Value	Z-Score
200	**-1.06**

300	**-0.71**
400	**-0.35**
600	**0.35**
1000	**1.77**

Z-Score Normalized Data

-1.06, -0.71, -0.35, 0.35, 1.77

Final Answers

(a) Min-Max Normalization (0–1)

Original	Normalized
200	0.000
300	0.125
400	0.250
600	0.500
1000	1.000

(b) Z-Score Normalization

Original	Z-Score
200	-1.06
300	-0.71
400	-0.35
600	0.35
1000	1.77

Q5. **Data Transformation in Online Shopping Behavior Analysis**

A. Equal-Frequency Binning

i. Divide the dataset into 3 bins

There are **10 records**, so divide them into **3 bins** with approximately equal records.

- **Bin 1:** 4 records
- **Bin 2:** 3 records
- **Bin 3:** 3 records

ii. Assign values to bins

Bin	Visit Duration (seconds)	Category
Bin 1	15, 30, 45, 60	Short Visit
Bin 2	120, 300, 600	Medium Visit
Bin 3	900, 1800, 3600	Long Visit

iii. Analysis

- Short visit durations are grouped together.
- Medium-duration visits form the second group.
- Long visits (900–3600 sec) are grouped together.
- Equal-frequency binning ensures each bin contains nearly the same number of records but the interval widths vary.

B. Z-Score Normalization

i. Calculate Mean

Data:

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600

Mean = 747 seconds

Step 2: Calculate Standard Deviation

X	$X - \mu$	$(X - \mu)^2$
15	-732	535824
30	-717	514089

45	-702	492804
60	-687	471969
120	-627	393129
300	-447	199809
600	-147	21609
900	153	23409
1800	1053	1108809
3600	2853	8139609

Variance

Standard Deviation

Standard Deviation = 1090.92 seconds

ii. Apply Z-Score Normalization

Formula

Visit Duration	Z-Score
15	-0.67
30	-0.66
45	-0.64
60	-0.63
120	-0.57

300	-0.41
600	-0.13
900	0.14
1800	0.97
3600	2.62

iii. Discussion

- Z-score standardizes data with **mean = 0** and **standard deviation = 1**.
- Extreme values like **1800** and **3600** receive high positive Z-scores.
- Outliers are **not removed**, but their distance from the mean is clearly shown.

C. Min-Max Normalization

i. Formula

Where

- Minimum = **15**
- Maximum = **3600**

Range

ii. Normalized Values

Visit Duration	Min-Max Normalized Value
15	0.000
30	0.004
45	0.008
60	0.013
120	0.029
300	0.080

600	0.163
900	0.247
1800	0.498
3600	1.000

iii. Comparison

Z-Score Normalization	**Min-Max Normalization**
Mean becomes **0**	Values lie between **0 and 1**
Standard deviation becomes **1**	Preserves relative positions
Outliers receive large positive/negative scores	Outliers compress the remaining values toward 0
Suitable for statistical analysis	Suitable for clustering and distance-based algorithms

Conclusion

- **Equal-frequency binning** groups customers into short, medium, and long visit durations with nearly equal numbers of records.
- **Z-score normalization** standardizes the data and highlights extreme values such as **1800** and **3600 seconds**.
- **Min-Max normalization** scales all visit durations to the **[0,1]** range, making the data suitable for clustering algorithms, though it is more sensitive to outliers than Z-score normalization.